gui libraries in assembly language or any library for simple ui just for colors

Thought for 1m 36s

There aren't many popular "GUI libraries written *in* assembly." Most assembly GUI programs do one of these:

1. **Call the OS GUI API directly** (Win32 on Windows, X11/Wayland on Linux)
2. **Link to a small C library** (SDL2 / raylib) and call its functions from assembly
3. If you only need **colors**, often the simplest "UI" is just **console colors** (ANSI / Windows Console)

Here are the best options, depending on what you mean by "simple UI just for colors":

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1) Just colored text/background (no real windows)

Windows console

Use **SetConsoleTextAttribute** to change text + background colors in the console.

Microsoft Learn

Linux / macOS terminal

Use **ANSI escape codes** (VT100). Fast, no GUI.

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2) A real window with a solid color (minimal GUI)

Windows (Win32 API + GDI)

This is the "native" way: create a window (CreateWindowEx) and paint it with **FillRect** (or a brush). Ciprian F. +1

If you're using **FASM**, its standard Win32 include packages help a lot (macros/imports).

Flat Assembler

Why this is good: no extra libraries, just `user32.dll + gdi32.dll`.

Linux (X11 / Xlib)

Create a window and set its background color with **XSetWindowBackground** (then clear/redraw). Tronche +1

Why this is good: very direct and lightweight on X11 systems.

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3) Easiest cross-platform "colored window" from assembly

SDL2

Create a window/renderer, then set a color and clear the screen with **SDL_SetRenderDrawColor** + **SDL_RenderClear**. SDL Wiki

Why this is good: works on Windows/Linux/macOS; you just link SDL2 and call its C ABI from assembly.

raylib

Super simple loop: open window + **ClearBackground(color)** style workflow (raylib's API is intentionally tiny). raylib +1

Why this is good: one of the simplest libraries for "window + colors + shapes".

Quick recommendation (if you tell me nothing else)

- If you only need **colors**: **console colors** (ANSI on Linux/macOS, SetConsoleTextAttribute on Windows). Microsoft Learn
- If you want an actual **window** and still keep it simple:
  - **SDL2** (most common) SDL Wiki
  - **raylib** (even simpler API) raylib +1
- If you want "pure OS, no extra libs":
  - **Win32 + GDI FillRect** on Windows Microsoft Learn +1
  - **Xlib** on Linux Tronche +1

If you tell me **your OS (Windows/Linux)** and **assembler (MASM/FASM/NASM)**, I'll point you to the cleanest setup and the exact libs to link (32-bit vs 64-bit matters for calling conventions).

Sources

i ma using windows 10 and want to make project in assembly language the project is too simple so let me

know what kind of libraries can i use for simple ui projects in assembly language x86

Thought for 25s >

On **Windows 10 + x86 assembly**, the most "simple UI" path is usually **not a special assembly GUI library**—it's calling **Windows' built-in GUI DLLs** (Win32 API). That keeps your project tiny and avoids extra dependencies.

Here are the practical options you can use:

1) Pure Win32 API (best for simple apps)

These are **system DLLs** you can call directly from assembly:

- **kernel32.dll**
  Basics: process start, memory, file I/O, `ExitProcess`, etc.
- **user32.dll**
  Real GUI: `CreateWindowEx`, message loop (`GetMessage/DispatchMessage`), buttons/edits/labels, `MessageBox`.
- **gdi32.dll**
  Simple drawing/colors: `CreateSolidBrush`, `FillRect`, `TextOut`, `Rectangle`, etc.

What you can build easily with just these:
A window + background color, text, buttons, input boxes, simple shapes.

Add-on: Common Controls (still built-in)

- **comctl32.dll**
  Nicer/standard controls: progress bar, trackbar, listview, toolbar, etc.
  You call `InitCommonControlsEx` and then create controls.

Add-on: GDI+ (images + better shapes)

- **gdiplus.dll**
  If you want PNG/JPG, anti-aliased shapes, gradients, etc.
  Slightly more setup, but still "standard Windows".

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2) "Easy graphics window" libraries (good if you only need colors/shapes)

These are **C libraries** but you can still use them from assembly by calling their exported functions (cdecl/stdcall depending on build):

- **SDL2**
  Super common: create window, clear screen with a color, draw rectangles, handle keyboard/mouse.
- **raylib**
  Even simpler API than SDL for beginners: open window, `ClearBackground`, draw shapes/text.

Why people choose these: you avoid Win32 message handling complexity, and you get a clean render loop.

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3) UI via Dialogs (fastest "forms-like" UI)

Still Win32, but you use a **resource script (.rc)** for UI layout (buttons, edits), then load it with:

- `DialogBoxParam / CreateDialogParam` (user32)

This is often the quickest way to make a "simple UI app" (like a small tool) without manually positioning every control in code.

My recommendation for your "too simple" project

If you want a normal Windows app (buttons, simple layout):

✅ **Win32 API only:** `kernel32 + user32 + gdi32`
(Optionally `comctl32` for nicer controls)

If you want just a colored window + simple drawing (no heavy UI):

✅ **SDL2** or **raylib** (simplest "window + color" experience)

Tooling note (important for x86)

Your choice depends a bit on assembler/linker:

- **MASM / UASM / JWasm**: easiest with Win32 import libs (`user32.lib`, `gdi32.lib`, etc.)
- **NASM**: also fine; many people link with **GoLink** or **MinGW ld**.

If you tell me which one you're using (**MASM / UASM / NASM / FASM**) I'll give you the *exact* minimal setup (includes + link command) and a tiny starter template for a colored window.

what about irvine

Irvine usually means **Irvine32 / Irvine64** (Kip Irvine's library). It's great for *learning assembly*, but it's **not a GUI library**.

What Irvine is good for

ChatGPT 5.2 Thinking

- Makes MASM homework/projects easier

What Irvine is *not* good for

- Creating real Windows UI (windows, buttons, menus)
- Drawing colored windows, forms, etc.

Colors with Irvine

You can do **console colors** (text/background in the console), but that's still **console**, not a window UI. Under the hood it uses Windows console APIs like `SetConsoleTextAttribute`.

If you want a "simple UI" on Windows 10 in x86 assembly

Pick one of these:

1. **Console UI (text + colors)**
   ✅ Irvine32 is perfect for this.
2. **Real window UI (buttons / window / background color)**
   ✅ Use **Win32 API** (`user32.dll` , `gdi32.dll`) directly in assembly.
3. **Simplest graphics window (clear screen with a color, draw shapes)**
   ✅ Use **raylib** or **SDL2** and call it from assembly.

If you tell me your setup:

- Are you using **MASM + Irvine32** in Visual Studio?

  I can suggest the easiest next step:
- **stay console** (Irvine colors), or
- switch to a **Win32 window template** (still MASM, no Irvine needed).

